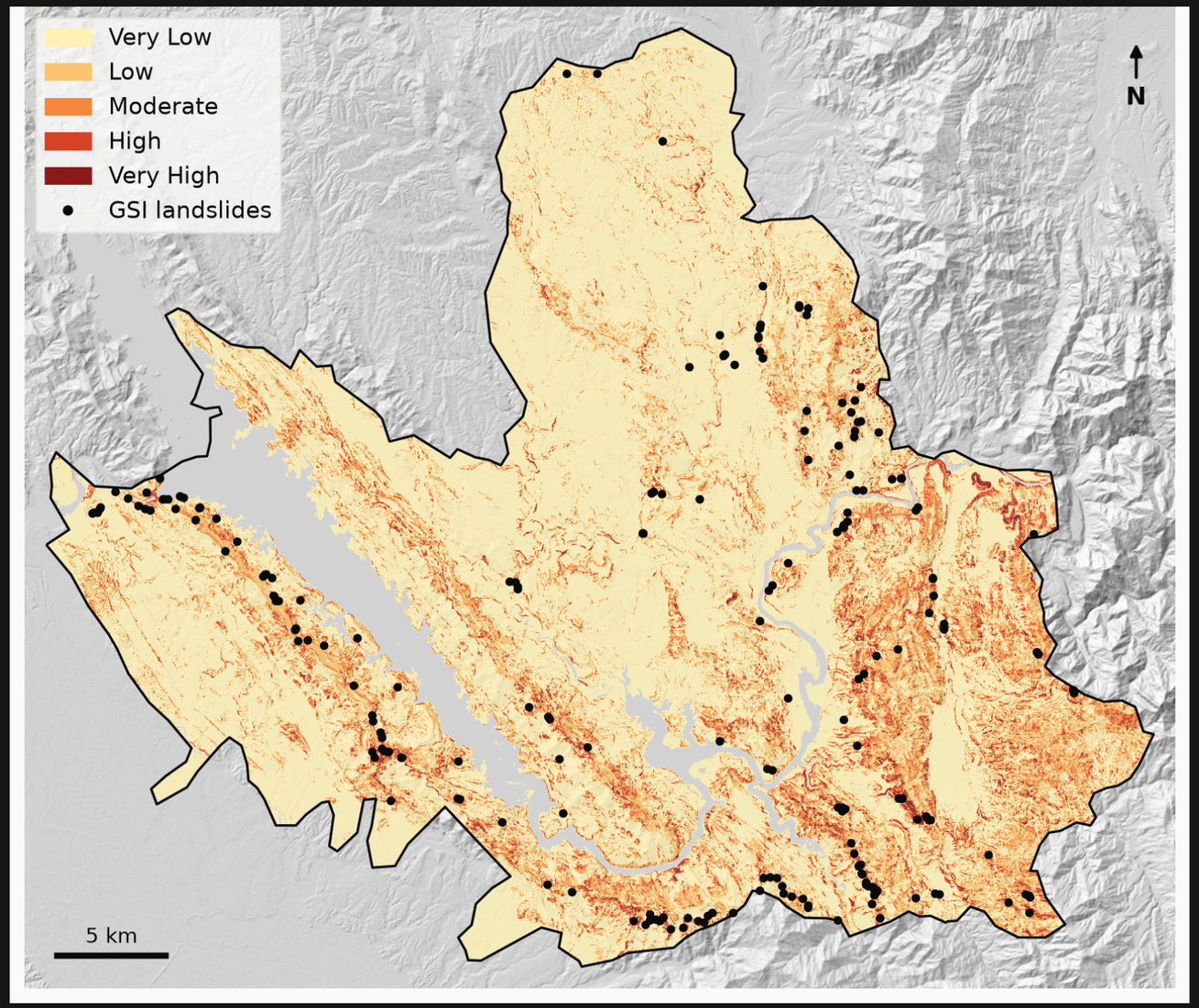
GEOSPATIAL ANALYSIS · MACHINE LEARNING

Landslide Susceptibility Analysis – Bilaspur

Mapping where slopes are likely to fail across Bilaspur district, Himachal Pradesh, by comparing an expert-weighted MCDA model with three data-driven models.



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October 2026

SUMMARY

A district-wide susceptibility map, validated on areas the model never saw

Using 219 mapped landslides, 563 stable points and six terrain and vegetation factors, five methods were compared with spatial cross-validation. Random Forest performed best and was used to map all 1.2 million 30 m cells in the district.

0.88

AUC of the terrain-only Random Forest under 2 km spatial cross-validation (0.92 with road distance)

+0.08

AUC gain over the expert-weighted **MCDA (AHP)** baseline (0.80), terrain-only models

4.5%

of the district falls in the **High / Very High** classes, which contain about 70% of mapped landslides

782

sample points: 219 GSI landslides and 563 stable points, each described by 6 factors

Data

GSI landslide inventory, Copernicus 30 m DEM, Sentinel-2 (297 scenes, 2019–24), ESA WorldCover, OpenStreetMap roads

Methods

Slope-only baseline, MCDA (AHP weights), Frequency Ratio, Logistic Regression, Random Forest

Tools

QGIS 3.44 · GRASS r.watershed · Google Earth Engine · Python (pandas, scikit-learn, rasterio)

BACKGROUND

From mapping one disaster to predicting the next

On **7 October 2025**, a slope failed onto a moving private bus near the Bhallu bridge in the Bhalughat area of Jhandutta, Bilaspur, killing at least **15 people**.

Phase 1 (2025): after the event

Compared Sentinel-2 imagery before and after the event using four spectral indices (NDVI, NDMI, NDWI, BSI). **NDVI change** traced the damaged area most sharply.

Phase 2 (this study): before the next one

Phase 1 shows *where a slide already happened*. This study asks **which parts of the district are prone to fail**, including places with no recorded slide yet.


Reuters World Business Markets Sustainability Legal Commentary Technology Investigations More My News

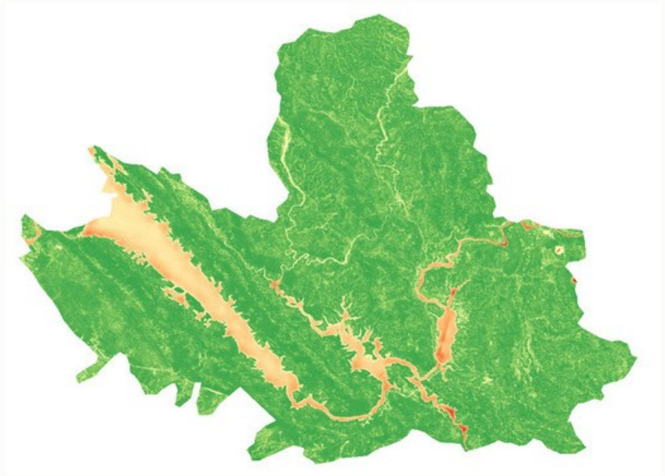
Death toll climbs to 15 after landslide hits bus in India's Himachal Pradesh

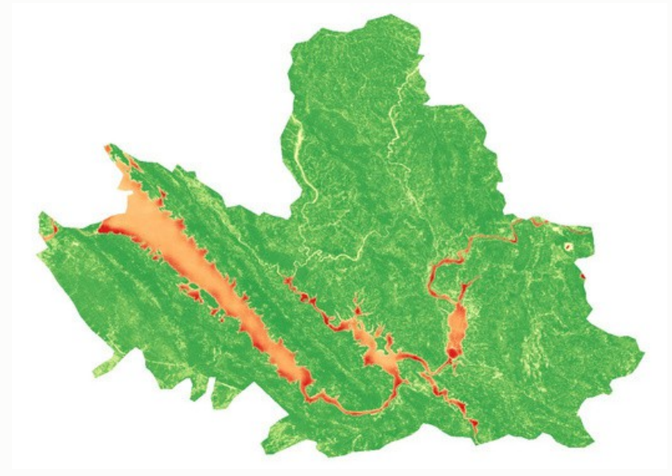
By Reuters

October 8, 2025 10:00 AM GMT+5:30 · Updated October 8, 2025

NEW DELHI, Oct 8 (Reuters) - The death toll from a landslide that struck a private bus in India's mountainous state of Himachal Pradesh has risen to 15, with one child still missing, officials and police said on Wednesday.

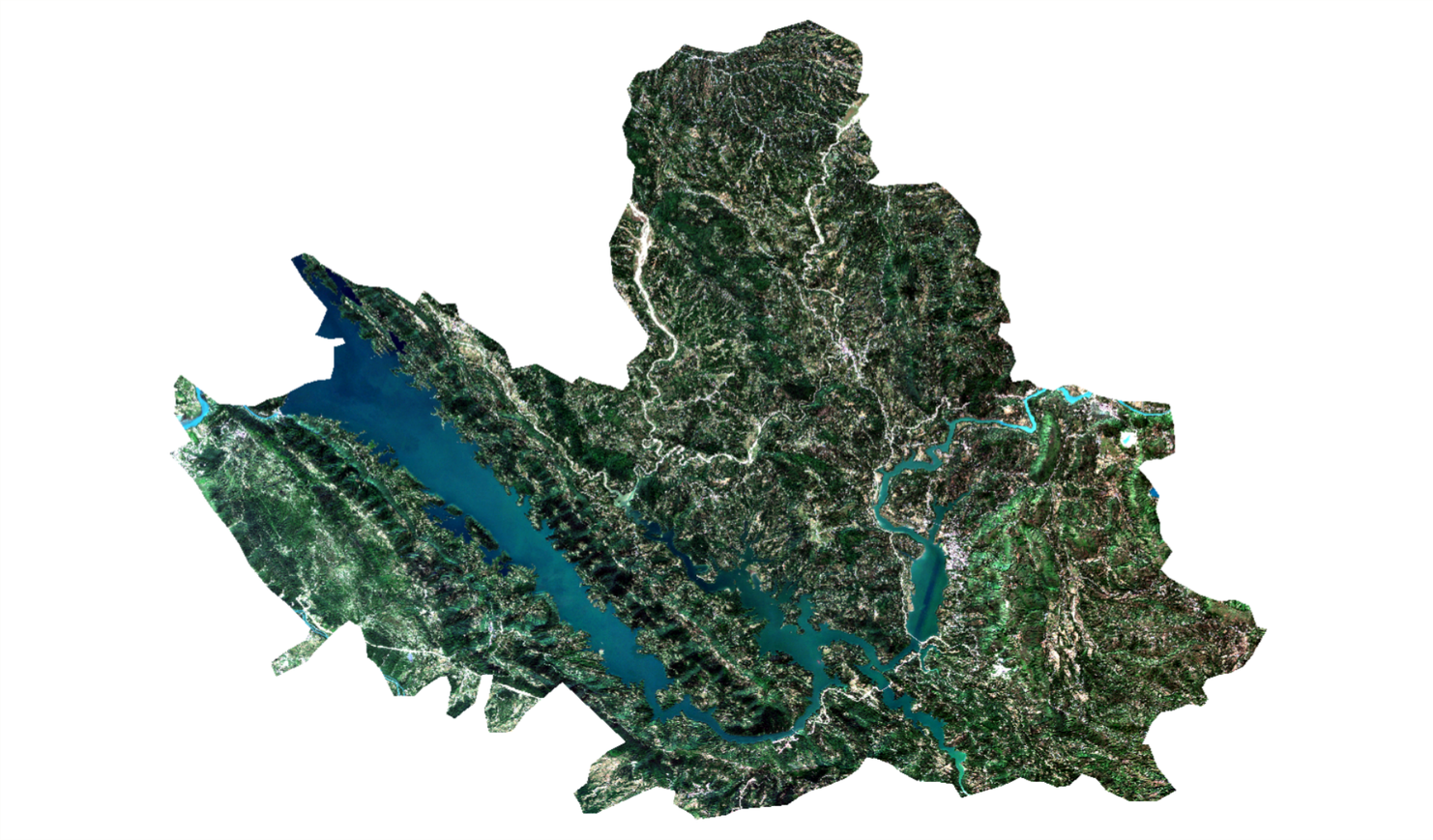
Reuters, 8 Oct 2025: report of the Bilaspur landslide on a bus.


Phase 1: NDVI before the event

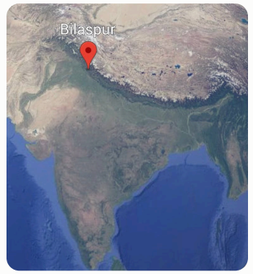

Phase 1: NDVI after the event

STUDY AREA

Bilaspur district, Lower Himalaya, Himachal Pradesh



Sentinel-2 true-colour composite clipped to the district. The Gobind Sagar reservoir (Bhakra Dam) runs north-west to south-east.



Located in the Shivalik and Lower Himalayan ranges, where monsoon rainfall, steep valley sides and road cutting make landslides a recurring hazard.

Area mapped	**≈ 1,166 km²**
Grid cells (30 m)	**1.30 million**
Cells on water (excluded)	**≈ 86 km²**
Typical elevation range	**≈ 400 – 1,200 m**
Coordinate system	**UTM 43N (EPSG:32643)**

The elevation range is the 2nd–98th percentile of the Copernicus DEM inside the district.

WORKFLOW

Six steps from raw data to a validated map

Each step was checked before moving on, and every model was scored on held-out areas, never on the points it learned from.

1

Inventory

GSI landslide points filtered to the district boundary and de-duplicated

QGIS · GADM

2

Stable points

Random points ≥ 500 m from any landslide, ≥ 200 m apart; water removed

QGIS

3

Factor rasters

Six factors on one aligned 30 m grid: terrain, vegetation, roads, streams

QGIS · GRASS · Earth Engine

4

Sampling

Factor values read at all 782 points into one table

Point Sampling Tool

5

Model comparison

Five methods scored by AUC with random and spatial cross-validation

Python · scikit-learn

6

Mapping

Best model applied to 1.2 M cells; 5 susceptibility classes

Python · rasterio

Key design decisions

Long-term NDVI (median, Oct–Nov 2019–24) so a single post-slide year cannot leak the answer · **Spatial blocks** so neighbouring points never sit in both training and test sets · **With / without road distance** to measure the effect of road-corridor survey bias

DATA

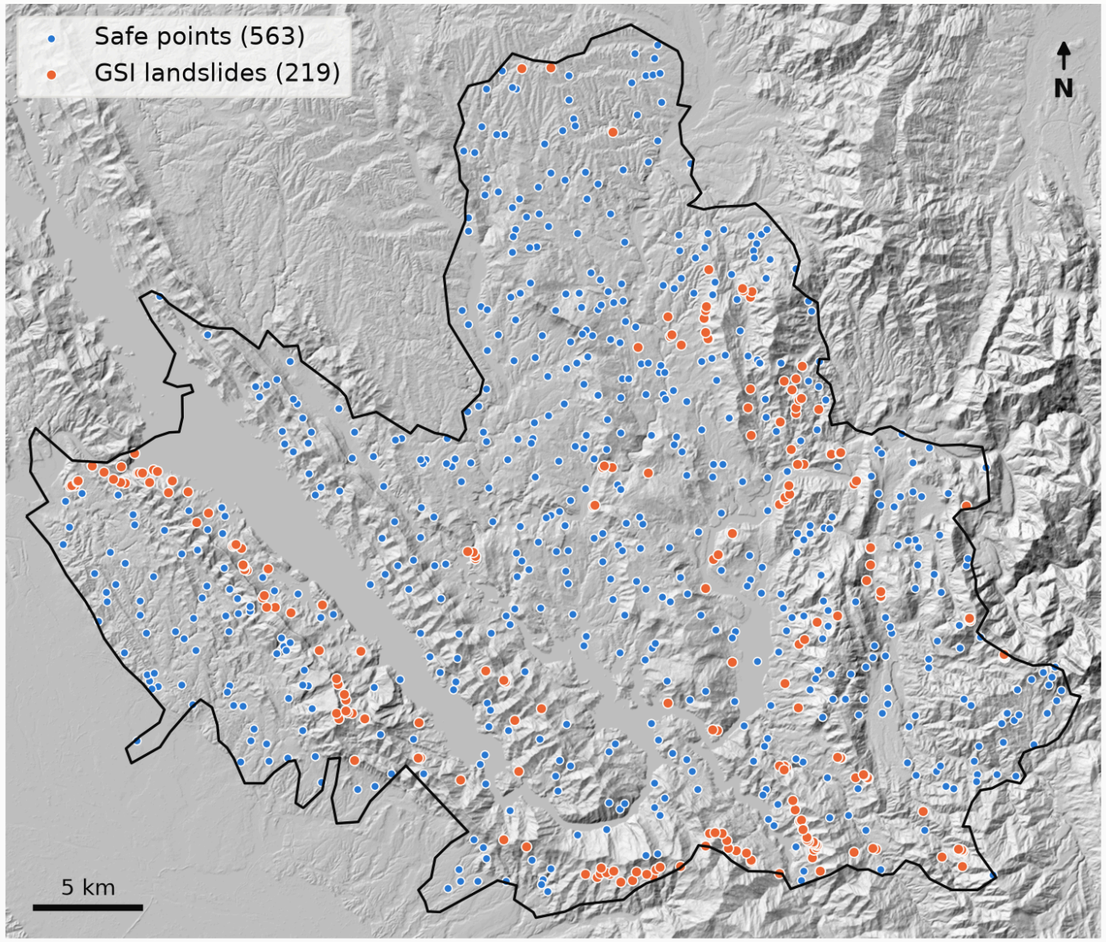
Open datasets, processed into six conditioning factors

DATASET	SOURCE	RESOLUTION	USED FOR
Landslide inventory	Geological Survey of India, via Mendeley Data (Chand, 2026)	Points	219 landslide locations (label 1)
Copernicus GLO-30 DEM	ESA, via OpenTopography	30 m	Elevation, slope, aspect, stream network
Sentinel-2 L2A	ESA Copernicus, via Google Earth Engine; Cloud Score+ masking	10 m → 30 m	NDVI: median of 297 post-monsoon scenes, 2019–24
ESA WorldCover 2021	ESA, via Google Earth Engine	10 m → 30 m	Water mask (class 80)
Road network	OpenStreetMap contributors (QuickOSM)	Vector	Distance to motorable roads (trunk to track, plus service)
District boundary	GADM 4.1	Vector	Study area and masking

Streams were derived from the DEM with GRASS r.watershed (threshold 1,000 cells ≈ 0.9 km² upstream area), because OpenStreetMap misses most small seasonal streams. All rasters were reprojected to UTM 43N and snapped to the DEM grid.

INVENTORY & SAMPLING

219 landslides and 563 stable points, filtered by location rather than label



Landslide points

6,539 GSI landslides, all of Himachal Pradesh

248 inside the Bilaspur boundary (spatial query)

219 after removing 29 duplicate locations

Stable (non-landslide) points

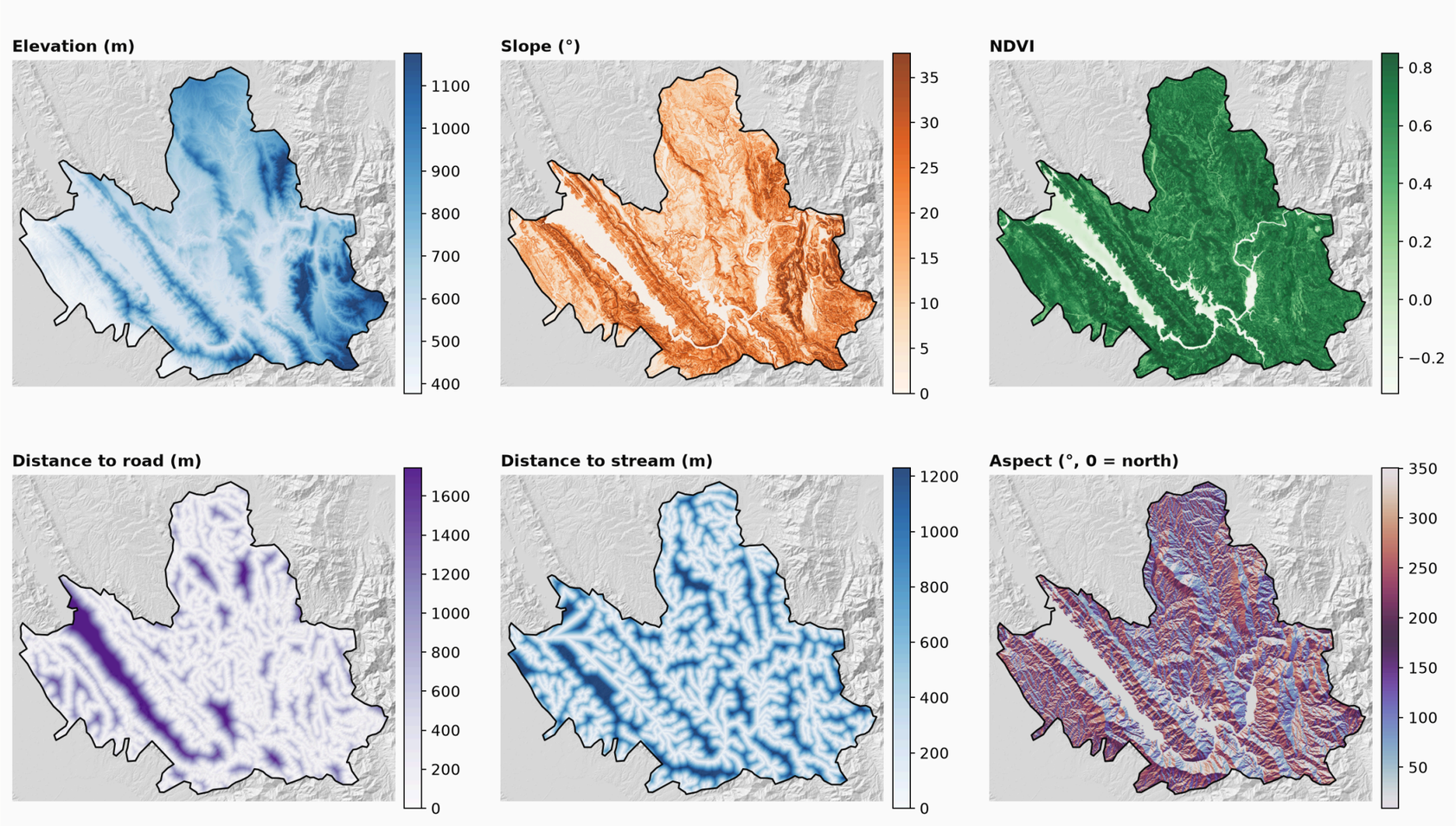
600 random points ≥ 500 m from any landslide, ≥ 200 m apart

563 after dropping 37 points on water

Why location, not labels: 21 records tagged “Bilaspur” plotted outside the district, so points were selected by spatial query against the boundary instead.

CONDITIONING FACTORS

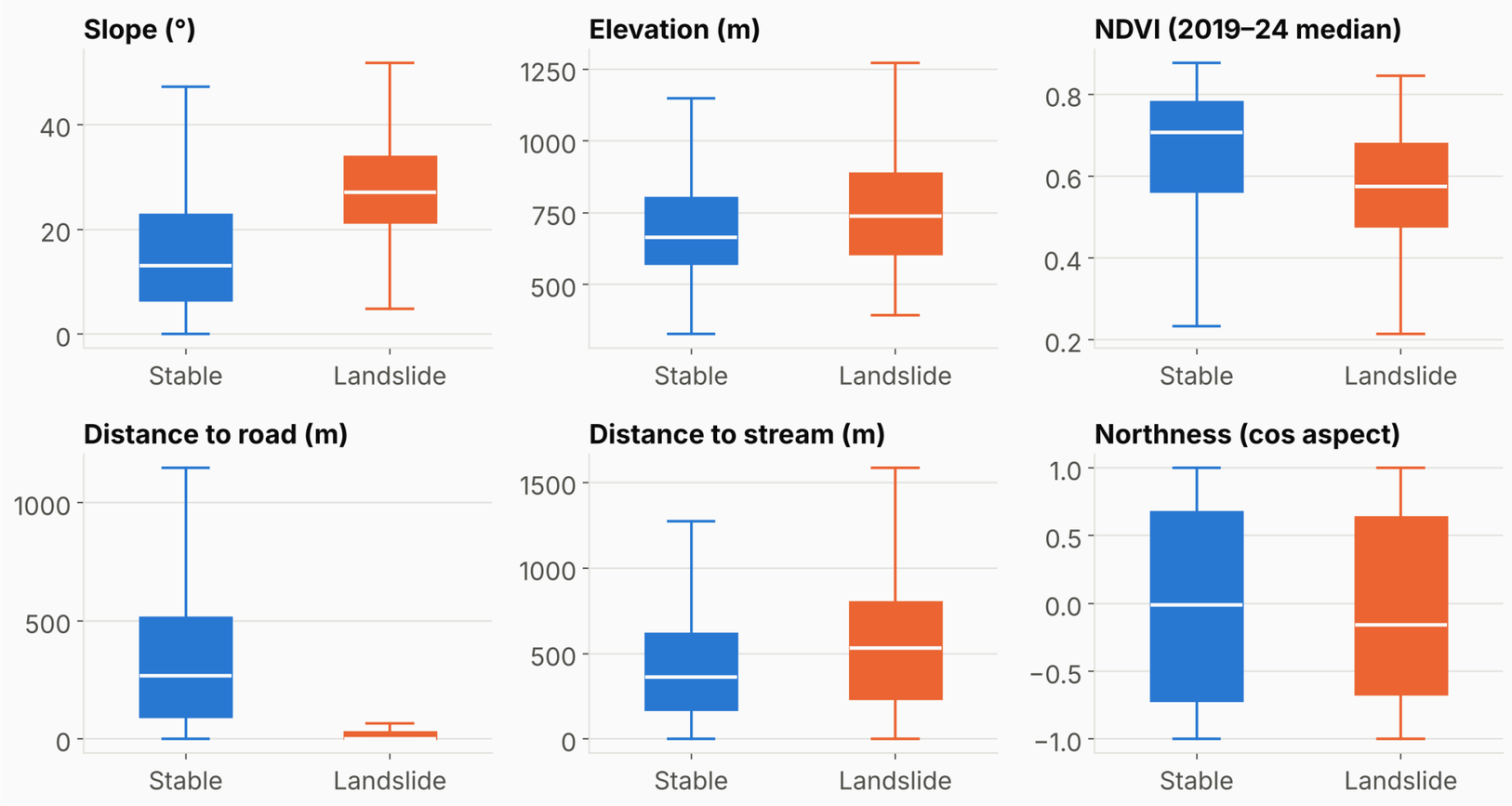
Six factors describing every 30 m cell



- **Elevation**: Copernicus DEM
- **Slope**: from the DEM, in degrees
- **Aspect**: converted to *northness* = cos(aspect) and *eastness* = sin(aspect), so 359° and 1° are treated as neighbours
- **NDVI**: vegetation density; roots bind soil
- **Distance to road**: road cuts steepen and unload slopes
- **Distance to stream**: streams erode the toe of slopes

EXPLORATORY ANALYSIS

Landslides sit on steeper, less vegetated ground, close to roads



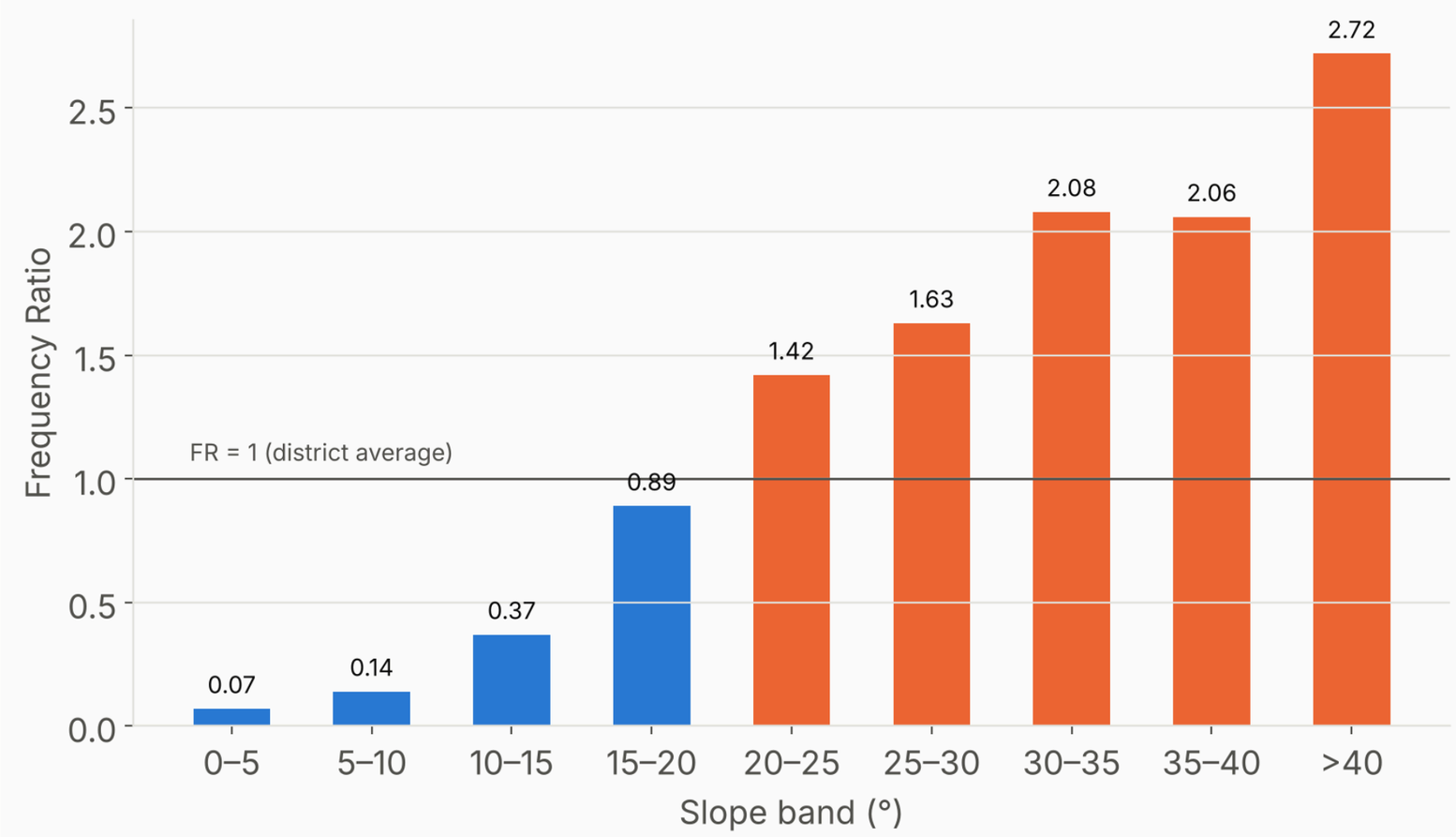
Stable Landslide

MEDIAN	STABLE	LANDSLIDE
Slope	13°	**27°**
NDVI	0.71	**0.57**
Distance to road	268 m	**0 m**
Distance to stream	362 m	**532 m**
Elevation	664 m	**738 m**

Aspect shows no difference on its own. Landslides lie *farther* from streams, probably because surveyed roads run along mid-slopes rather than valley floors.

FREQUENCY RATIO

Landslide frequency rises steadily with slope, with no single safe threshold



How FR works

FR = (share of landslides in a band) ÷ (share of all points in that band). **FR > 1** means landslides are over-represented. Summing a point's FR values across all factors gives its risk score.

- Slopes above 40° hold landslides at **2.7×** the district average; slopes below 5° at **0.07×**.
- A single “never safe above X°” cut-off would either miss many slides or flag almost half the district, so graded classes are used instead.

METHODS & VALIDATION

Five methods, from expert judgement to machine learning

METHOD	LEARNS FROM DATA?	HOW IT SCORES A LOCATION
Slope only	No	Steeper = riskier (baseline)
MCDA (AHP)	No	Each factor scored 1–5 by fixed rules, weighted by AHP literature weights (slope 0.40, road 0.16, stream 0.15, NDVI 0.15, elevation 0.07, aspect 0.06)
Frequency Ratio	Yes, bivariate	Sum of the FR values of each factor band
Logistic Regression	Yes, linear	Weighted sum through a sigmoid; standardised inputs, log-transformed distances
Random Forest	Yes, non-linear	500 trees, min. 5 points per leaf; captures interactions such as “steep *and* bare”

Metric: AUC

The probability that a random landslide point scores higher than a random stable point. 0.5 is chance; above 0.8 is good; above 0.9 is excellent.

Spatial cross-validation

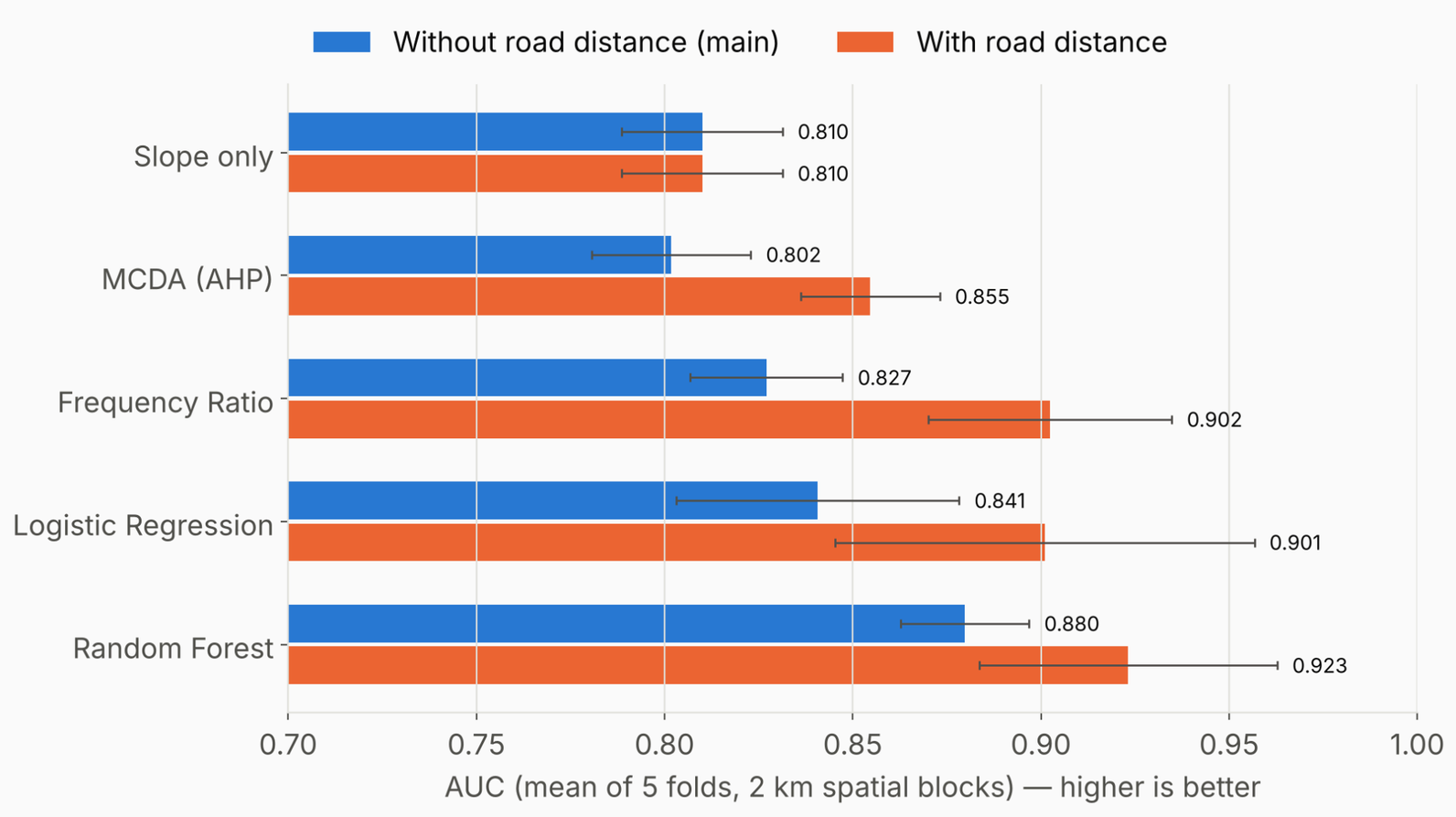
Half of all landslides have another within 233 m, so random splits leak information. Points were grouped into **2 km and 5 km blocks**, and whole blocks were held out together (5 folds).

Bias check

54% of landslides sit on a road cell because GSI surveys follow roads. Every model was run **with and without** road distance.

RESULTS

Random Forest leads, beating the MCDA baseline with and without road data



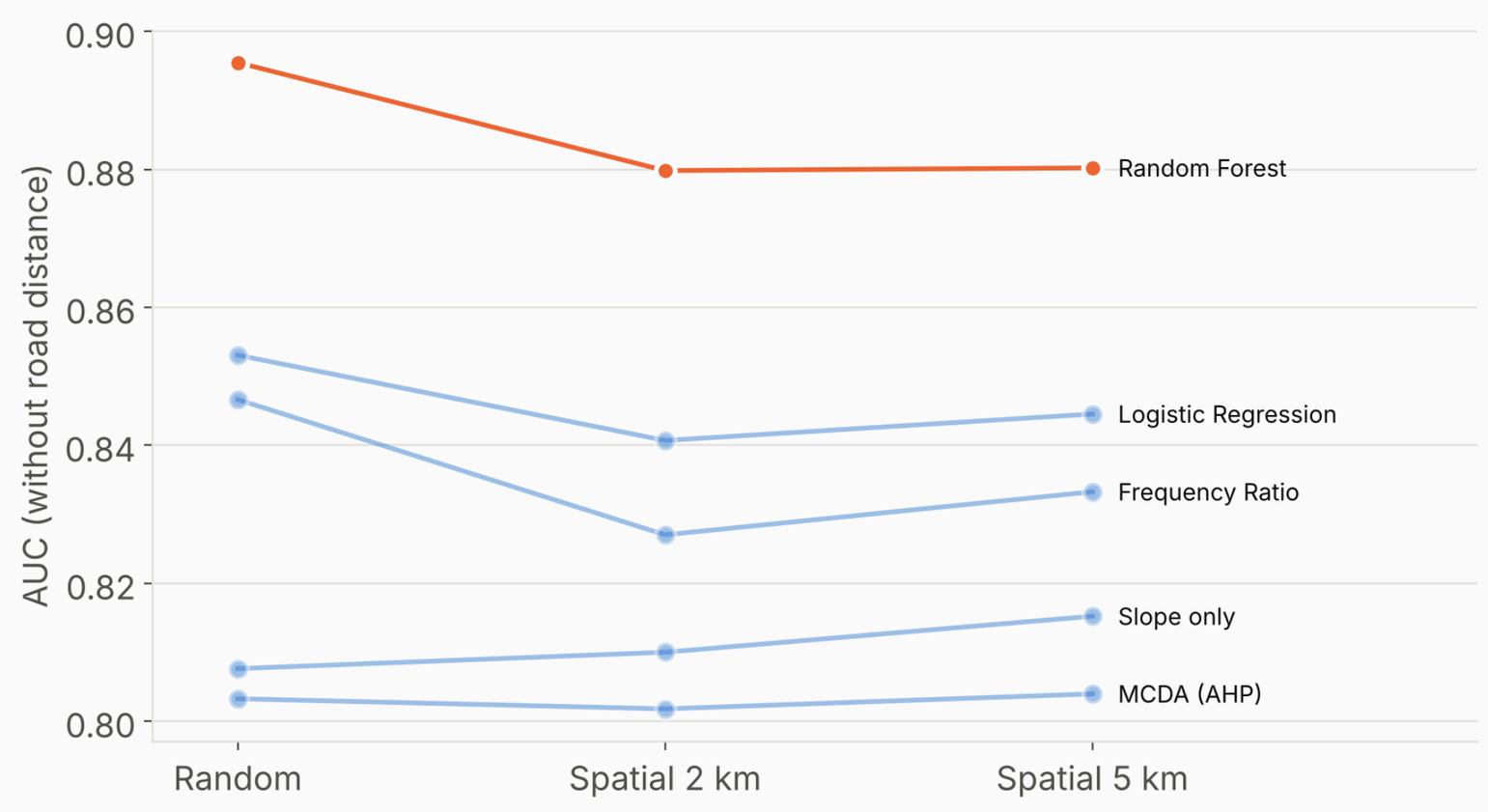
0.880

Random Forest, terrain only, 2 km spatial CV, vs **0.802** for MCDA (AHP)

- **All data-driven methods beat MCDA**. Without roads, the expert-weighted model is no better than slope alone (0.80 vs 0.81).
- **Random Forest gains most where the bias is removed** (+0.04 over Logistic Regression), which points to real interactions between factors.
- Road distance adds about 0.04–0.08 AUC to every multi-factor model. Part of this is real road-cut instability and part is survey bias.

ROBUSTNESS

Scores barely drop when models are tested on unseen areas

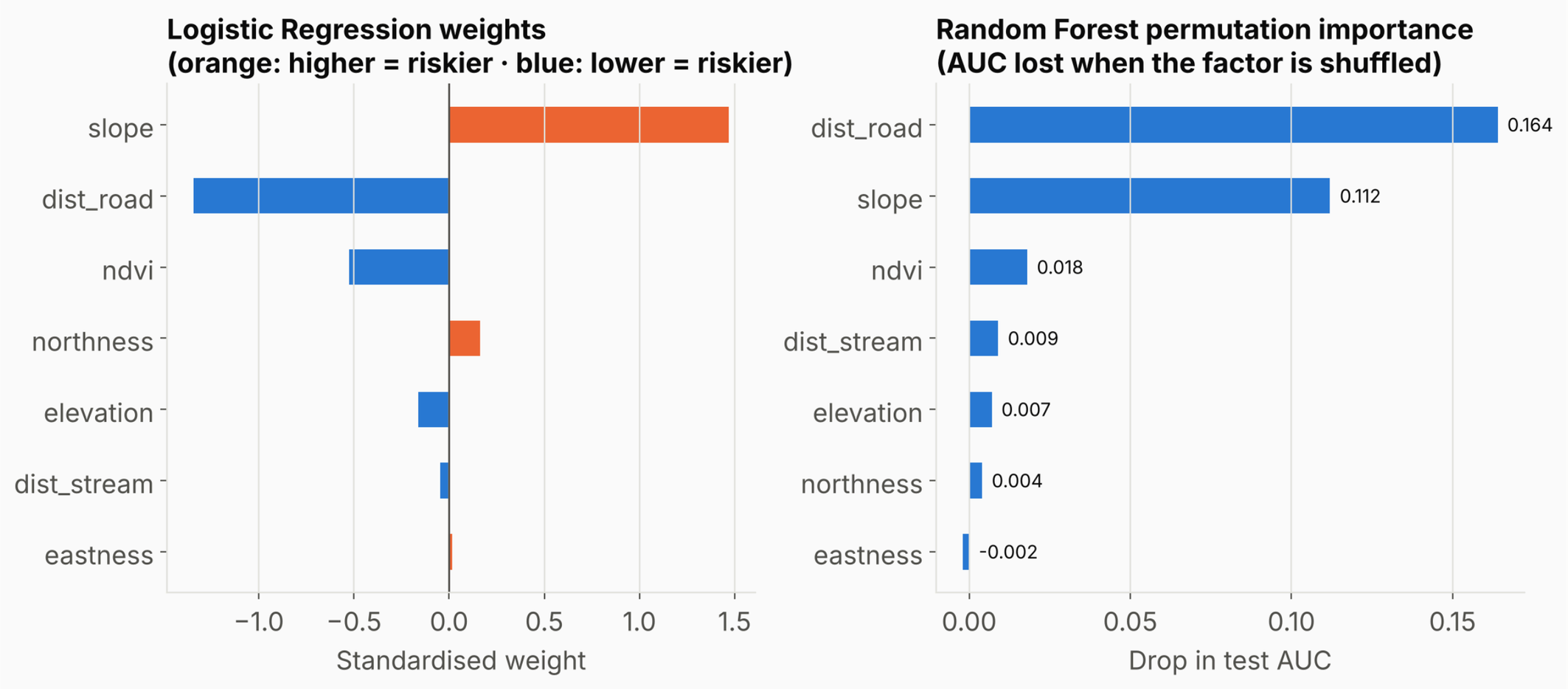


AUC, TERRAIN ONLY	RANDOM	2 KM	5 KM
Slope only	0.808	0.810	0.815
MCDA (AHP)	0.803	0.802	0.804
Frequency Ratio	0.847	0.827	0.833
Logistic Regr.	0.853	0.841	0.845
Random Forest	**0.895**	**0.880**	**0.880**

Learned models lose only **0.01–0.02** AUC under spatial blocking, and the ranking holds at both block sizes, so the models learned general terrain rules rather than memorising locations. Untrained methods (slope, MCDA) stay flat, which is a built-in sanity check.

WHAT DRIVES SUSCEPTIBILITY

Slope, road proximity and vegetation do almost all the work



Both models agree on the top three: slope, distance to road and NDVI. Aspect and stream distance add almost nothing in Bilaspur.

Elevation flips sign once slope is controlled for. Higher ground is steeper, so elevation on its own mostly repeats slope's signal.

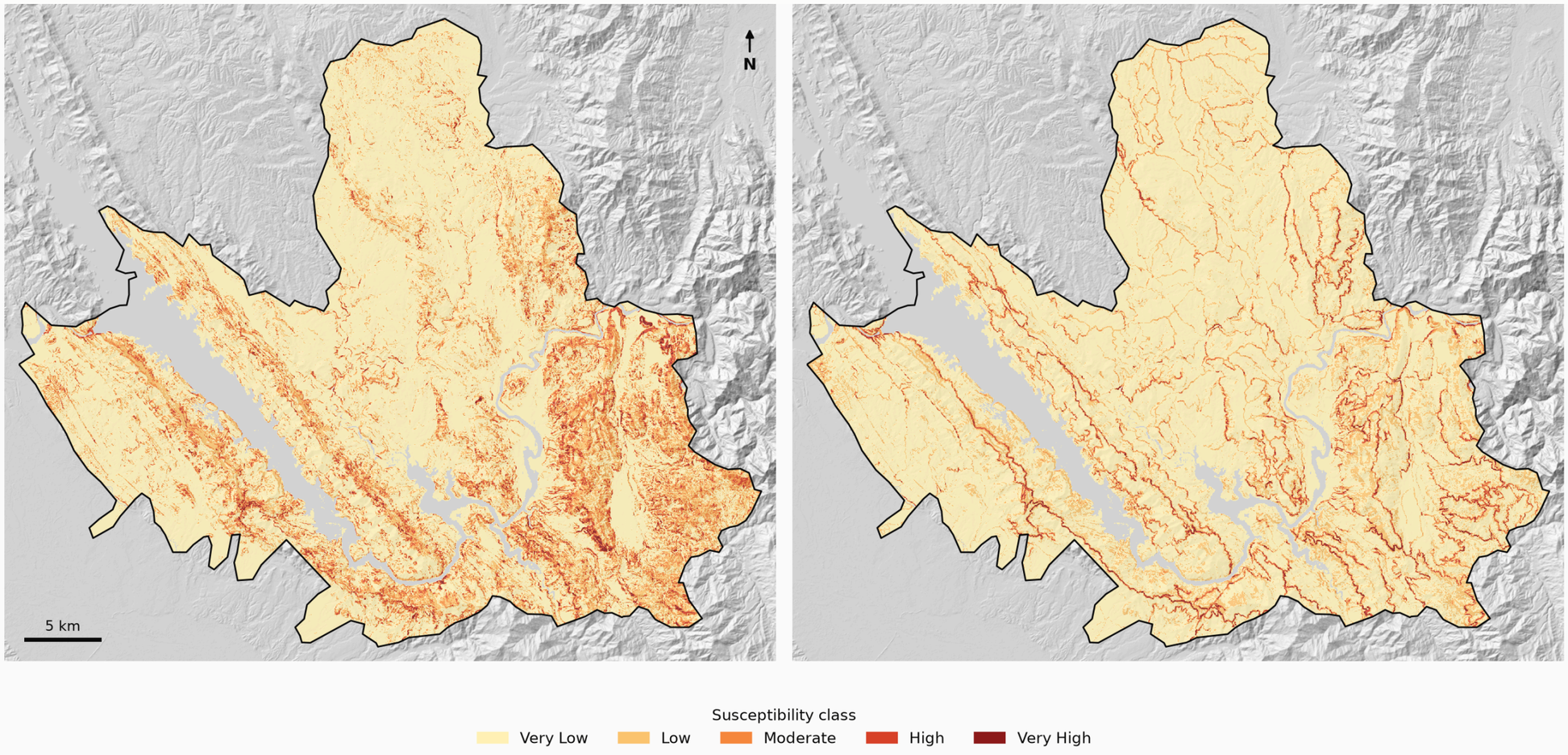
Permutation importance was measured on held-out folds (AUC lost when a factor is shuffled), which avoids the bias of tree-based importance towards continuous factors.

SUSCEPTIBILITY MAPS

Terrain-only map vs the road-informed map

A · Main **Random Forest, terrain only** (AUC 0.88). Risk follows steep, sparsely vegetated valley sides, concentrated in the south and east.

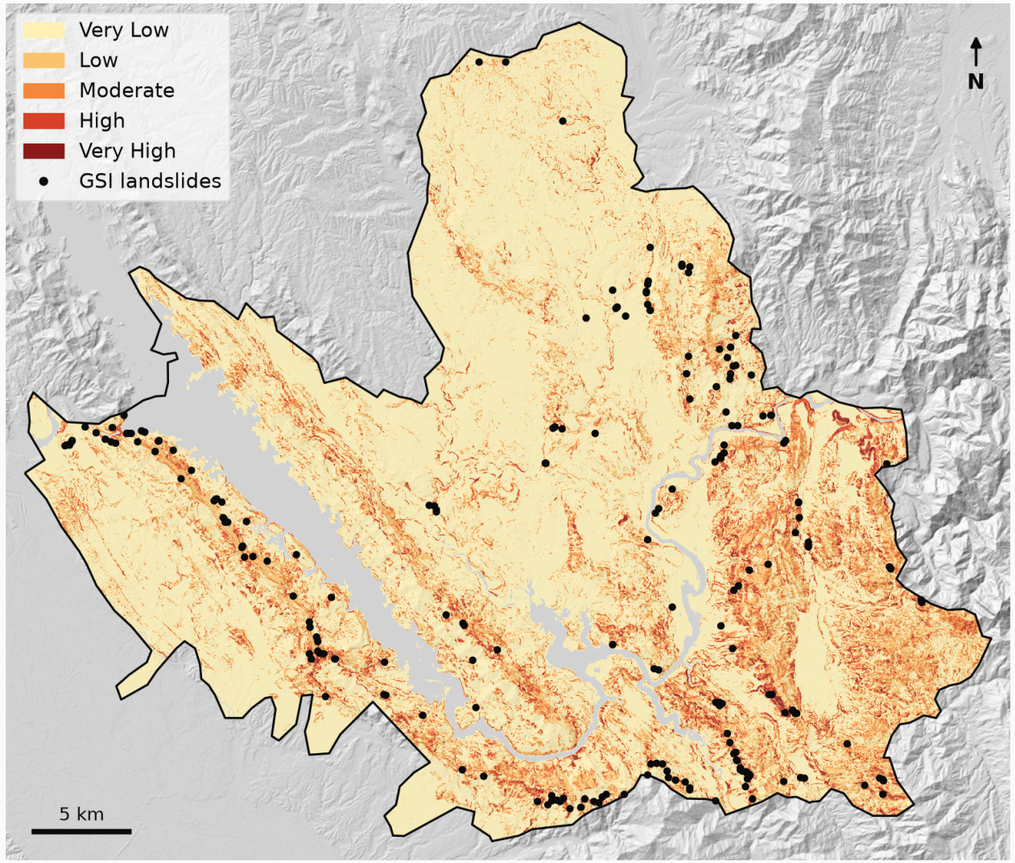
B · Secondary **Random Forest with road distance** (AUC 0.92). Risk traces the road network, showing both road-cut hazard and where surveys were done.

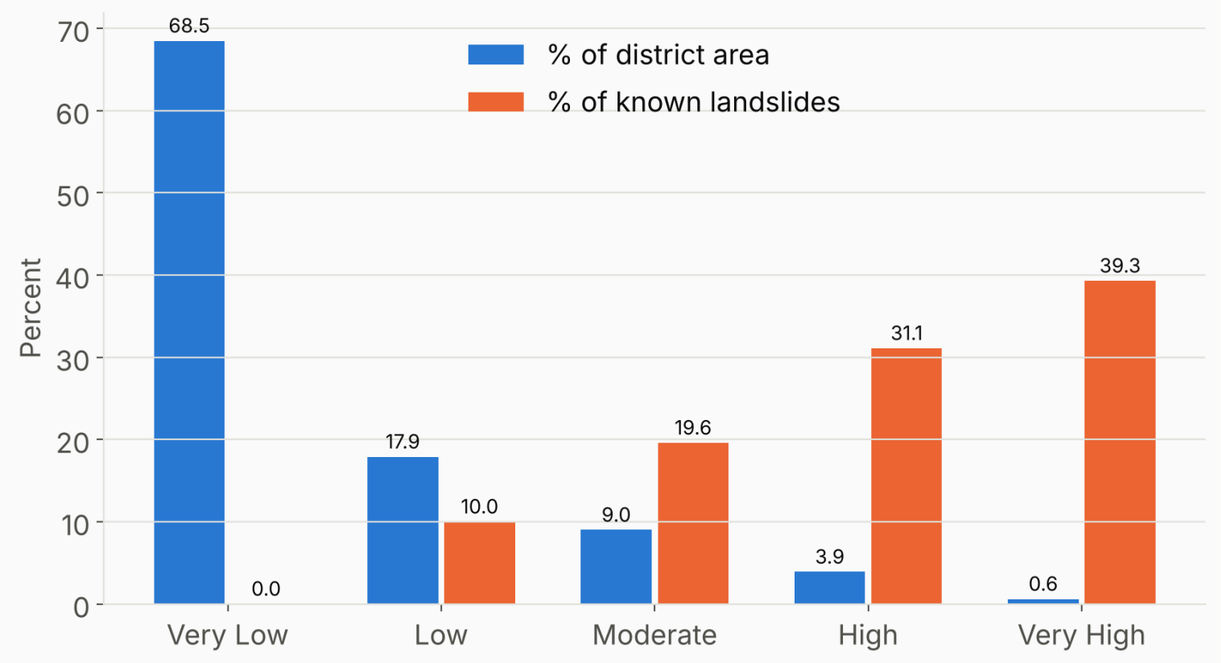


Classes are five equal intervals between 0 and the map's highest predicted probability. Grey areas are water.

FINAL MAP

4.5% of the district holds about 70% of mapped landslides





CLASS	PROBABILITY	% OF AREA	% OF LANDSLIDES
Very High	0.77 – 0.96	0.6	**39.3**
High	0.58 – 0.77	3.9	**31.1**
Very Low	0 – 0.19	68.5	**0.0**

Class shares use the training landslides, so they show how risk is concentrated, not independent accuracy. Accuracy is the spatial-CV AUC.

LIMITATIONS

What the model can and cannot claim

Road-corridor bias

GSI mapped mostly along roads (54% of landslides fall on a road cell). The terrain-only map is the main result for this reason.

Missing dates

Only 7.5% of records are dated, so NDVI is a 2019–24 median rather than the pre-event value. This limits, but does not remove, leakage from post-slide vegetation loss.

30 m resolution

The DEM averages out fresh road cuts a few metres high, which are a common trigger for small slides.

Susceptibility, not timing

The map shows where slopes are prone to fail, not when. Rainfall triggers and lithology are not yet included.

Label noise

Some “stable” points may be unrecorded slides, especially away from roads.

Event check not used

The exact initiation point of the Oct 2025 slide could not be located reliably, so it was not used as a validation point.

CONCLUSIONS

Findings and next steps

Findings

- A Random Forest using only terrain and vegetation reaches **AUC 0.88** on unseen 2 km blocks, **+0.08** over the MCDA (AHP) baseline and +0.07 over slope alone.
- **Slope, road proximity and vegetation** dominate. Aspect and stream distance add little in Bilaspur.
- Expert AHP weights over-weight factors that turned out to be weak here, so **local data should calibrate the weights**.
- High-risk terrain is concentrated: **4.5%** of the district holds about 70% of mapped landslides.

Next steps

- Add **rainfall** (CHIRPS) and **lithology** (GSI geology) as factors.
- Validate on an **independent, post-2025 inventory**, ideally mapped away from roads.
- Train across **neighbouring districts** (Mandi, Solan, Hamirpur) to test transferability.
- Use higher-resolution terrain data to capture **road-cut slopes**.

0.88

Headline: a terrain-only Random Forest, validated on held-out 2 km blocks, outperforms the expert-weighted MCDA model (0.80) and a slope-only baseline (0.81), and produces a district-wide map at 30 m resolution.

REFERENCES & CREDITS

Data sources and acknowledgements

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Software

QGIS 3.44 with GRASS GIS, QuickOSM and Point Sampling Tool · Google Earth Engine · Python 3.13: pandas, NumPy, scikit-learn, rasterio, GeoPandas, matplotlib

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